# Erin Avllazagaj

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## Research Interests

My broad research interests cover data-driven malware analysis and automatic exploit generation. Specifically, in my recent PhD work I have analyzed executions of malware traces in the real world to derive lessons for future behavior based detection systems. I'm currently interested in automatic exploit generation for heap based exploits. My participation in various CTF competitions and my work Octavian have been major influences in the new direction for my PhD.

## Education

**University of Maryland** — College Park, MD

***Ph.D.*** *student in Electrical and Computer Engineering, Advisor: Prof. Tudor Dumitraş* — *Aug. 2018 – Present*

**Bilkent University** — Ankara, Turkey

***B.S.*** *degree in Computer Science* — *Aug. 2014 – May 2018*

- Scholarly paper: "Privacy-Related Consequences of Turkish Citizen Database Leak" **(INFER'16)**
  - Advisors: *Prof. Erman Ayday, Prof. Ercüment Çiçek*
- Senior Project: "Andlit"
  - Advisor: *Prof. Ibrahim Körpeoğlu*
  - The project is a social network of individual face recognition classifiers. Designed to help the visually impaired recognise their environment, the friends in front of them and the friends of their friends. The project was called a "third eye" by various Turkish media agencies [1*] [2] [3] [4] [5] including daily newspapers.

## Professional Experience

**Maryland Cybersecurity Center** | *Research Intern* | *Maryland, USA* — June 2017 – August 2017

- I worked with Octavian Suciu to investigate code resuse of exploits in the wild. In this work I read many papers and used their static analysis techniques to measure closeness of code snippets parsed with Joern.

**4S (contractor for HavelSan)** | *Red Teaming* | *Ankara, Turkey* — July 2016 – August 2016

- I was tasked with testing a Unity based game for security issues. The game was a red-blue team simulation. The testing was important due to the client-server architecture and the security implications the server would impose at HavelSan. Without access to the source code I used tools from DevXUnityUnpacker, to Cheat Engine and IDA Pro.

## Publications

### Conferences

**E. Avllazagaj**, Z. Zhu, L. Bilge, D. Balzarotti, T. Dumitraş, "When Malware Changed Its Mind: An Empirical Study of Variable Program Behaviors in the Real World", In 30th USENIX Security Symposium (USENIX Security 21), 2021

### Workshops

**E. Avllazagaj**, E. Ayday, E. Çiçek, "Privacy-Related Consequences of Turkish Citizen Database Leak", In International Workshop on Inference & Privacy in a Hyperconnected World (INFER' 16), 2016

### Posters

O. Suciu, **E. Avllazagaj**, T. Dumitraş, "The Secret Life of Pwns: Characterizing and Predicting Exploit Weaponization", In Conference on Applied Machine Learning for Information Security (CAMLIS' 19), 2019

## Honors and Awards

### Fellowships

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|---|---|
| Clark Doctoral Fellowship GRA/GTA support (4 years) | 2018 – 2022 |
| Clark Doctoral Fellowship stipend (first 2 semesters) | 2018 – 2019 |
| Clark Doctoral Fellowship travel grant (4 years) | 2018 – 2022 |
| Bilkent University Full Tuition+Accommodation scholarship (all 4 years) | 2014 – 2018 |
| Mehmet Akif High School Tuition scholarship (all 3 years) | 2011 – 2014 |

### Awards

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|---|---|
| 2nd place in Informatics Olympiad of Tirana | 2013 |

## Services

### Program Committee

[Artifact Evaluation] Usenix 2021 — 2021

[Artifact Evaluation] PoPETS (Accepted papers from PETS 2021) — 2021

### External reviewer

Network and Distributed System Security Symposium (NDSS) — 2019 – 2020

USENIX Security Symposium (USENIX) — 2018 – 2019

International Symposium on Research in Attacks, Intrusions and Defenses (RAID) — 2018, 2019

ACM Conference on Computer and Communications Security (CCS) — 2018 – 2020

## Projects

### Analysis of malware behavior in the wild

*Paper status* — *Accepted to USENIX Security 2021*

**Abstract**: Behavioral program analysis is widely used for understanding malware behavior, for creating rule-based detectors, and for clustering samples into malware families. However, this approach is ineffective when the behavior of individual samples changes across different executions, owing to environment sensitivity, evasive techniques or time variability. The models created and tested using such traces do not account for the broad range of behaviors observed in the wild, and may result in a false sense of security.

In this project, we conduct the first quantitative analysis of behavioral variability in malware, PUP and benign samples, using a novel dataset of 7.6M execution traces, recorded in 5.4M real hosts from 113 countries. We analyze program behaviors at multiple granularities, and we show how they change across hosts and across time. We then analyze the invariant parts of the malware behaviors, and we show how this affects the effectiveness of malware detection using a common class of behavioral rules.

### Automatic generation of slab exploit primitives

*Paper status* — *In progress*

**Abstract**: Slab allocators are being used in Linux kernel to facilitate allocation of structures with minimal fragmentation. However, unlike user-land allocators, slab allocators store little to no in-place metadata, making them a less promising target for exploitation. Therefore, it requires significant manual efforts to transform off-by-one bug into a successful exploit.

In this project, we propose an automated approach to discover exploitation primitives to transform a seemingly useless bug to a more useful unsafe heap state. Through a layout-guided we demonstrate a method that can find exploit primitives for various implementations of the slab allocator. We then translate the sequence of kmalloc's and kfree's to the system calls provided by the kernel.

## Teaching Experience

### Bilkent University — Ankara, Turkey

*Lab Teaching Assistant* — *Spring 2016*

**Class**: CS 223: Digital Design

**Tasks**: Help students with solving the lab problems using System Verilog. I also taught the procedure of designing using Xilinx Vivado Design Suite and the differences of System Verilog to Verilog.